CFD Analysis of Deflagration-to-Detonation Transition for Bird Deflection

COURSE CODE AND COURSE

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CERTIFICATE

Certified that this Seminar Report titled CFD Analysis of Deflagration-to-Detonation Transition for Bird Deflection by Darsh Butala is approved by me for submission. Certified further that, to the best of my knowledge, the report represents work carried out by the student.

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ABSTRACT

Bird strikes continue to pose a major risk to aviation safety and result in significant financial losses worldwide. Conventional bird deterrent systems, while widely used, often suffer from a decline in effectiveness due to habituation—where birds gradually adapt to repetitive stimuli and cease to respond. This report investigates the scientific principles underlying the Deflagration-to-Detonation Transition (DDT), as the foundation for developing a more effective avian deterrence system. By utilizing the rapid energy release and high acoustic intensity associated with a controlled detonation, such a device can provide a strong, non-habituating stimulus to disperse birds from critical zones like airports. The report discusses the fundamental differences between deflagration and detonation, explains the transition mechanism between these combustion regimes, and explores the potential applications, engineering design aspects, and safety considerations for implementing a DDT-based bird scaring device.

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Chapter 1

INTRODUCTION

Bird activity in and around airports presents an enduring challenge to aviation safety. Collisions between birds and aircraft—commonly known as bird strikes—can lead to severe engine damage, structural failure, and, in extreme cases, catastrophic accidents. The global economic impact of bird strikes, including repair costs and operational disruptions, is estimated to amount to billions of dollars annually.

A wide range of bird control methods has been implemented over the years, including habitat modification, bio-acoustic deterrents (such as distress calls), pyrotechnics, and propane cannons. Although these methods initially achieve successful results, their long-term effectiveness diminishes as birds become accustomed to the repetitive nature of the deterrent stimuli. In particular, propane cannons—commonly used for producing loud acoustic blasts through simple deflagration—often fail to maintain effectiveness because the subsonic combustion lacks sufficient shock intensity to continually startle or deter birds.

This report explores the feasibility of developing a bird scaring system based on the Deflagration-to-Detonation Transition (DDT) phenomenon. The concept involves generating a controlled detonation. The resulting supersonic shockwave and intense acoustic output would serve as a more startling and unpredictable deterrent, thereby minimizing the likelihood of habituation. Such a system has the potential to significantly enhance airfield safety and reliability compared to conventional methods.

Chapter 2

DEFINING COMBUSTION REGIMES: DEFLAGRATION AND DETONATION

Combustion phenomena can be broadly categorized into two regimes—deflagration and detonation—based on the speed of the combustion wave relative to the local speed of sound and the dominant physical mechanisms responsible for propagation.

2.1 DEFLAGRATION

Deflagration refers to a subsonic mode of combustion in which the flame front propagates through the unburned mixture primarily via thermal conduction and molecular diffusion. Energy transfer from the high-temperature combustion products to the unreacted gases ahead of the flame causes the latter to heat up and ignite progressively.

Propagation Speed: Ranges from a few centimeters per second to several hundred meters per second, but remains below the speed of sound in the medium.

Pressure Characteristics: Results in a modest and gradual rise in pressure across the flame front.

Dominant Mechanism: Controlled primarily by heat transfer and diffusion processes.

Typical Examples: Burning of a candle, a flash fire, or the acoustic discharge from a conventional propane cannon.

While deflagration-based devices are simple and relatively safe, their limited energy release and subsonic combustion nature restrict the magnitude of the resulting acoustic impact, thereby reducing long-term deterrent effectiveness.

2.2 DETONATION

Detonation represents a more energetic and violent mode of combustion that propagates at supersonic speeds, typically between 1,500 and 2,500 m/s for common fuel–air mixtures. In contrast to deflagration, detonation involves a shock wave–reaction zone coupling mechanism. The leading shock wave compresses and heats the unreacted mixture

to extremely high pressures and temperatures, triggering autoignition almost instantaneously behind the front.

Propagation Speed: Supersonic, typically 1.5–2.5 km/s.

Pressure Characteristics: Produces a sharp and significant pressure spike—often several orders of magnitude greater than that of a deflagration.

Dominant Mechanism: Driven by compressive heating caused by the shock wave, followed by rapid chemical energy release that sustains the propagation.

Typical Examples: High-explosive detonations and nuclear explosions.

This highly energetic process generates a powerful shockwave accompanied by an intense acoustic signature, which forms the basis of the proposed DDT-based bird scaring device. The combination of unpredictability and intensity makes detonation-based systems particularly effective at producing strong aversive responses in birds, while maintaining the potential for controlled and safe operation through precise engineering design.

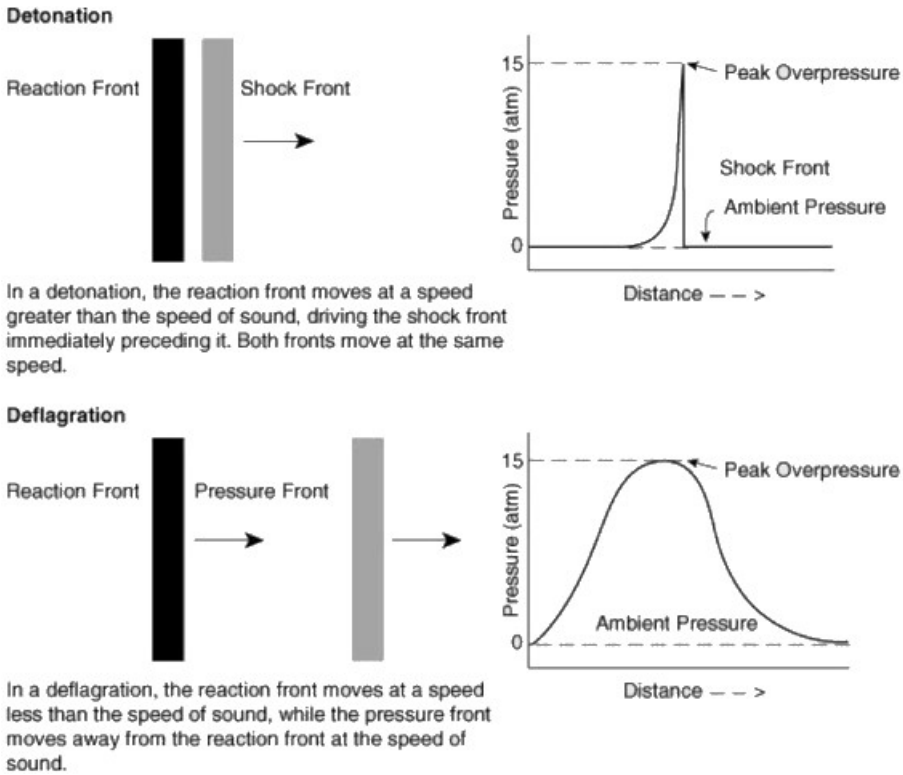

Figure 2.1: Comparison of detonation versus deflagration combustion.

Chapter 3

THERMODYNAMIC AND GAS DYNAMIC FOUNDATIONS OF COMBUSTION WAVES

A thorough understanding of the Deflagration-to-Detonation Transition (DDT) requires first establishing the thermodynamic and gas dynamic framework that governs steady detonations. The theoretical description of a one-dimensional, steady detonation wave has evolved from an idealized thermodynamic perspective to a more sophisticated model that accounts for finite-rate chemical kinetics and real-gas effects.

In essence, the study of detonation waves combines both thermodynamics—to define possible equilibrium states—and fluid dynamics—to describe how the wave propagates through a compressible medium. The fundamental conservation principles that govern these transitions are encapsulated in the Rankine–Hugoniot relations, which serve as the cornerstone for analyzing the behavior of shock and detonation waves.

3.1 THE HUGONIOT CURVE: LOCUS OF POSSIBLE THERMODYNAMIC STATES

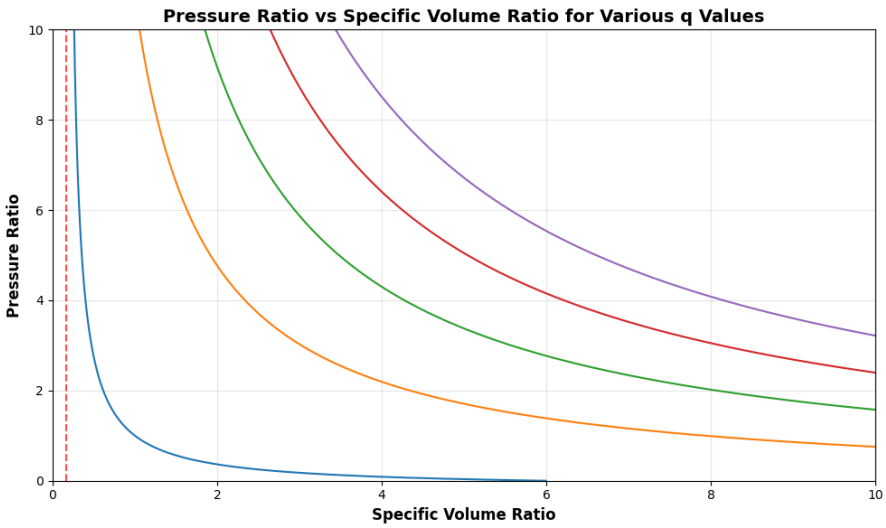

Figure 3.1: Pressure Ratio vs. Specific Volume Ratio as a function of heat release (q).

The Hugoniot curve is a graphical representation of all possible final thermodynamic

states that satisfy the energy conservation condition for a given initial state. It is typically plotted in the pressure–specific volume (p–v) plane.

$$\left(\frac{P_2}{P_1} + \frac{\gamma - 1}{\gamma + 1}\right)\left(\frac{v_1}{v_2} - \frac{\gamma - 1}{\gamma + 1}\right) = \frac{4\gamma}{(\gamma + 1)^2} + \frac{2\gamma q}{a_1^2}\left(\frac{\gamma - 1}{\gamma + 1}\right)$$

3.2 THE RAYLEIGH LINE: MECHANICAL CONSTRAINT FOR A SPECIFIC WAVE VELOCITY

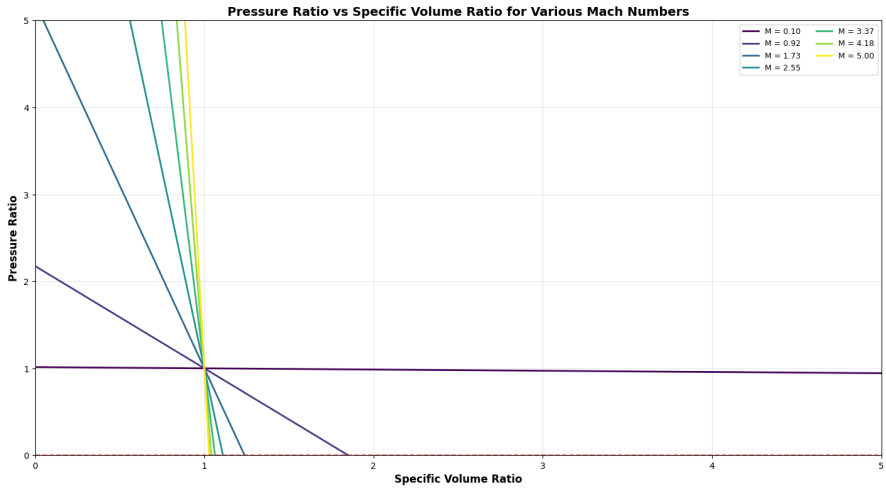

Figure 3.2: Pressure Ratio vs. Specific Volume Ratio as a function of Mach No. (M).

While the Hugoniot curve defines all thermodynamically possible post-wave states, it does not by itself determine the velocity of the propagating wave. The relationship between the mechanical properties of the flow and the wave velocity is provided by the Rayleigh line, derived by combining only the mass and momentum conservation equations. This equation defines a straight line in the p–v diagram, whose slope depends on the wave velocity.

$$\frac{P_2}{P_1} - 1 = \gamma M_1^2 \left(1 - \frac{v_1}{v_2}\right)$$

A physically valid final state must satisfy all three conservation laws—mass, momentum, and energy—and therefore must correspond to the intersection of the Rayleigh line with the Hugoniot curve.

This geometric relationship between the Hugoniot curve and the Rayleigh line provides the foundation for understanding how a deflagration wave can accelerate and eventually transition into a detonation wave under suitable physical and chemical conditions.

Chapter 4

THE CHAPMAN–JOUGUET CONDITION: THE SELECTION RULE FOR STABLE DETONATION

Experimental studies consistently reveal that an unconfined, self-sustaining detonation in a given reactive mixture propagates at a unique and reproducible velocity. This observation suggests that, although multiple post-detonation states may satisfy the Rankine–Hugoniot relations, nature selects only one specific state that is both dynamically and thermodynamically stable.

The Chapman–Jouguet (C–J) condition provides this unique selection criterion. It defines the precise condition under which a detonation wave can propagate in a steady and self-sustaining manner, without requiring any external energy input. The C–J condition effectively identifies a single, physically realizable point on the Hugoniot curve, corresponding to a specific detonation velocity known as the Chapman–Jouguet velocity.

4.1 MATHEMATICAL FORMULATION: THE TANGENCY CONDITION

Chapman and Jouguet independently proposed that a stable detonation wave corresponds to the limiting case in which the Rayleigh line—which represents the mechanical constraint for a given mass flux—is tangent to the reactive Hugoniot curve in the p–v plane.

This point of tangency, known as the Chapman–Jouguet point, signifies a unique detonation state that satisfies all three conservation laws—mass, momentum, and energy—while maintaining dynamic stability. Mathematically, this condition implies that the slope of the Rayleigh line equals the slope of the Hugoniot curve at the point of intersection:

$$\left(\frac{dp}{dv}\right)_{\text{Rayleigh}} = \left(\frac{dp}{dv}\right)_{\text{Hugoniot}}$$

At this point, both thermodynamic and mechanical constraints are simultaneously fulfilled, yielding a unique solution for the detonation velocity and corresponding state properties.

4.2 PHYSICAL INTERPRETATION: THE SONIC FLOW CRITERION

Beyond the mathematical definition, the C–J condition possesses a clear physical interpretation. At the Chapman–Jouguet point, the flow of combustion products immediately behind the detonation front is sonic relative to the wave front. This means that the velocity of the products in the reference frame fixed to the detonation wave equals the local speed of sound in the burned gases:

$$M_2 = \frac{u_2}{c_2} = 1$$

where:

- M_2 = Mach number of the burned gases (in the wave-fixed frame),
- u_2 = flow velocity of burned gases relative to the detonation front,
- c_2 = equilibrium speed of sound in the combustion products.

The sonic condition implies that no pressure disturbances or information can propagate upstream from the burned gases to influence the reaction zone or the leading shock. Consequently, the detonation is self-sustaining, relying entirely on the energy released by the chemical reactions occurring just behind the front to drive the shock wave forward.

This criterion ensures the stability and independence of the detonation front—an essential characteristic of real, experimentally observed detonations.

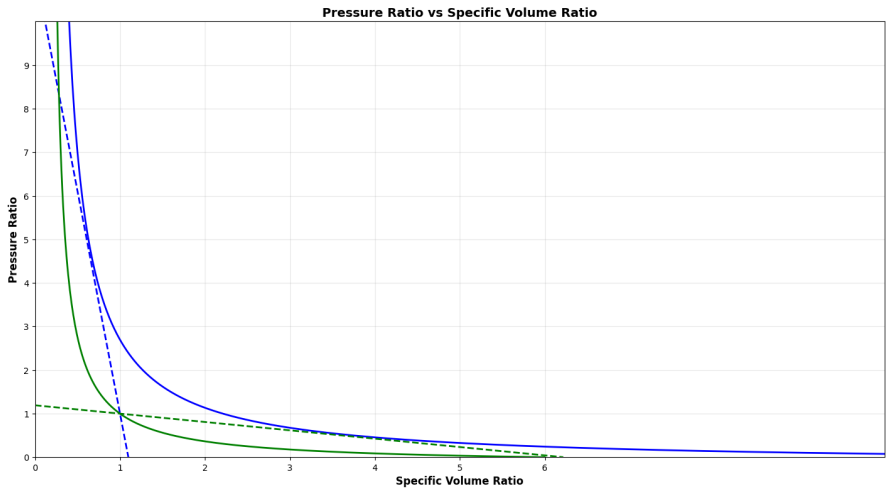

Figure 4.1: Pressure Ratio versus Specific Volume Ratio for Detonation Waves.

Chapter 5

CLASSIFICATION OF ONE-DIMENSIONAL COMBUSTION WAVES

The pressure–specific volume (p–v) diagram, which features the Hugoniot curve along with several Rayleigh lines emerging from the initial state point (p_1, v_1), provides a comprehensive framework for visualizing and classifying all possible one-dimensional, steady combustion waves.

The Chapman–Jouguet points—one on the upper branch and one on the lower branch of the Hugoniot curve—serve as critical boundaries separating different combustion regimes. The upper branch corresponds to detonation waves, characterized by supersonic propagation, while the lower branch represents deflagration waves, where combustion occurs subsonically.

5.1 THE UPPER HUGONIOT BRANCH: DETONATION REGIMES

The upper branch of the Hugoniot curve corresponds to supersonic combustion waves in which both pressure and density increase across the wave front. Depending on the slope of the corresponding Rayleigh line, several detonation regimes can be identified:

STRONG (OVERDRIVEN) DETONATION

This regime corresponds to Rayleigh lines steeper than the C–J tangent, implying a wave velocity greater than the Chapman–Jouguet condition.

CHAPMAN–JOUGUET DETONATION

This is the unique, self-propagating detonation state where the Rayleigh line is tangent to the Hugoniot curve. The downstream flow is sonic ($M_2 = 1$), preventing any feedback from the products to the front. This is the physically realizable and stable detonation most commonly observed in experiments.

WEAK DETONATION

Occurs when the Rayleigh line has a shallower slope than the C–J tangent but remains steeper than the line connecting the initial state to the minimum of the Hugoniot curve.

5.2 THE LOWER HUGONIOT BRANCH: DEFLAGRATION REGIMES

The lower branch of the Hugoniot curve represents subsonic combustion waves, or deflagrations, where the pressure decreases and the specific volume increases across the flame front. These regimes are similarly categorized based on the slope of the Rayleigh line relative to the lower C–J point:

WEAK DEFLAGRATION

Corresponds to shallow-sloped Rayleigh lines, producing subsonic flow ($M_2 < 1$) behind the flame front. This regime describes ordinary laminar flames, where the propagation speed is governed primarily by thermal conduction and molecular diffusion rather than gas dynamic effects.

CHAPMAN–JOUGUET DEFLAGRATION

Represents the tangency point on the lower branch where the Rayleigh line just touches the Hugoniot curve. The downstream flow is sonic ($M_2 = 1$), corresponding to the maximum possible steady deflagration velocity. This state may be approached in highly turbulent or confined flames, though rarely achieved in practice.

STRONG DEFLAGRATION

Corresponds to steeper Rayleigh lines between the C–J tangent and the vertical. The disturbances could travel upstream faster than the flame itself—an outcome that violates the second law of thermodynamics. Hence, strong deflagrations are physically unrealizable.

The combined p–v diagram analysis thus offers an elegant and unified framework to visualize the continuum of combustion phenomena—from slow, subsonic deflagrations to fast, supersonic detonations. The Chapman–Jouguet condition serves as the fundamental boundary between stable and unstable regimes, dictating the attainable steady states of one-dimensional combustion waves.

Chapter 6

THE ZELDOVICH–VON NEUMANN–DÖRING (ZND) MODEL OF DETONATION STRUCTURE

While the Chapman–Jouguet (C–J) theory successfully predicts the velocity of a stable detonation, it models the detonation front as an infinitesimally thin discontinuity and does not capture the internal processes that occur within the wave. To address this limitation, the Zeldovich–von Neumann–Döring (ZND) model introduces a more realistic, one-dimensional framework that resolves the finite structure of the detonation wave. This model divides the detonation process into two sequential regions: a leading shock front and a finite-rate chemical reaction zone that follows it.

6.1 STRUCTURE OF THE ZND WAVE

The ZND model describes the detonation process as a sequence of three distinct stages:

LEADING SHOCK WAVE

The process begins with a strong, non-reactive shock front that instantaneously compresses and heats the unreacted explosive mixture. This adiabatic compression raises both pressure and temperature to extremely high values, producing a transient state known as the *von Neumann spike*. At this point, the pressure reaches its maximum value, considerably exceeding the final C–J pressure, while the chemical reactions have yet to commence.

CHEMICAL REACTION ZONE

The elevated thermodynamic conditions at the von Neumann spike trigger chemical reactions within the mixture. This reaction zone possesses a finite thickness, determined by the reaction kinetics of the explosive system. It typically consists of two subregions:

- *Induction zone:* Radical species form through chain-branching reactions with minimal heat release.

- *Main reaction zone:* Exothermic recombination reactions release most of the chemical energy.

As the reactions proceed, the pressure gradually decreases from the von Neumann spike toward the final equilibrium value.

TERMINATION AT THE CHAPMAN–JOUGUET (C–J) STATE

The reaction zone ends at the C–J plane, where the chemical reactions are essentially complete. At this point, the flow of the detonation products relative to the wave front becomes sonic, fulfilling the C–J condition. Beyond this point, the combustion products expand further in a rarefaction region.

The ZND model thus bridges the gap between macroscopic gas-dynamic behavior and microscopic chemical kinetics. By introducing a finite reaction zone, it establishes intrinsic length and time scales—namely, the reaction zone thickness and the chemical induction time—that are governed by the specific reaction rates of the explosive mixture. This provides a theoretical foundation for studying detonation initiation and propagation mechanisms.

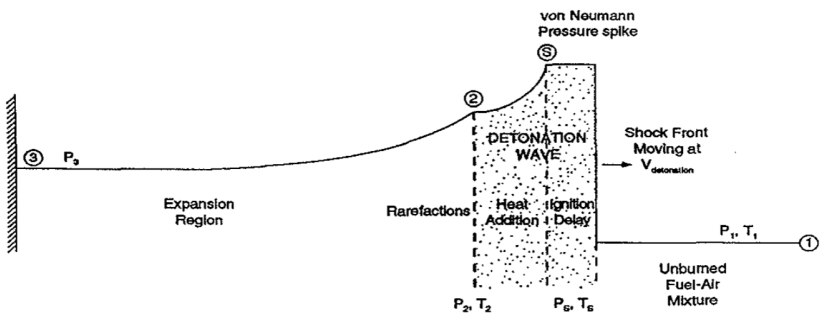

Figure 6.1: ZND Detonation Wave Structure Profile, showing the von Neumann pressure spike, reaction zone, and subsequent expansion waves.

6.2 THERMODYNAMIC AND FLOW PROPERTY PROFILES

The progression of the ZND wave is reflected in the variation of key thermodynamic and fluid-dynamic quantities across the structure:

PRESSURE PROFILE

The pressure undergoes a sharp rise at the shock front, reaching its maximum at the von Neumann spike. It then gradually decreases throughout the reaction zone as chemical energy is converted into kinetic and internal energy of the gases. At the C–J state, the pressure stabilizes before declining further through an unsteady expansion wave.

TEMPERATURE PROFILE

Similar to pressure, the temperature experiences a sudden increase across the shock front. Within the reaction zone, it continues to rise as exothermic reactions proceed, typically reaching a peak before stabilizing at the final equilibrium value. In some systems with a distinct endothermic induction phase, a temporary temperature plateau or dip may precede the main temperature rise.

DENSITY PROFILE

The density increases abruptly at the shock, attaining a maximum at the von Neumann spike due to compression. As chemical reactions release heat and the gases expand, the density steadily decreases through the reaction zone until it reaches the equilibrium density at the C–J point.

6.3 THERMODYNAMIC REPRESENTATION ON THE P–v DIAGRAM

The thermodynamic evolution of the ZND detonation can be conveniently represented on a pressure–specific volume (P–v) diagram. The Rayleigh line, derived from the conservation of mass and momentum, connects the initial unreacted state to the possible post-shock states.
In this framework:

- The shock compression moves the system from the initial point to the von Neumann spike, which lies on the inert Hugoniot (representing non-reactive compression states).
- The subsequent chemical reaction drives the state along the Rayleigh line as energy is released, ultimately reaching the equilibrium Hugoniot, which represents the fully reacted products.
- A stable Chapman–Jouguet detonation occurs at the unique condition where the Rayleigh line is tangent to the equilibrium Hugoniot, signifying the sonic flow condition at the end of the reaction zone.

6.4 FUNDAMENTAL ASSUMPTIONS AND LIMITATIONS

The predictive capability of the ZND model is based on several simplifying assumptions, which also define its inherent limitations:

- **One-Dimensional Flow:** The model assumes that all variables vary only along the direction of wave propagation, neglecting multi-dimensional effects.

- **Steady-State Propagation:** The internal structure of the detonation is considered time-invariant, with a constant propagation velocity.
- **Simplified Chemistry:** The chemical reaction mechanism is often reduced to a single-step, irreversible exothermic reaction in an ideal gas mixture.

While these assumptions make the model analytically tractable, they limit its applicability to real-world detonations, which are inherently multi-dimensional, transient, and chemically complex. Despite these simplifications, the ZND model remains a foundational framework for understanding the internal structure and dynamics of detonation waves.

Chapter 7

MECHANISMS OF DEFLAGRATION-TO-DETONATION TRANSITION (DDT)

The transition from a subsonic deflagration to a supersonic detonation is a complex, multi-stage phenomenon rather than an instantaneous process. It begins with a period of rapid flame acceleration, during which the combustion front interacts dynamically with its surroundings to produce strong pressure waves and localized regions of high temperature. These conditions, in turn, act as precursors to the eventual formation of a self-sustaining detonation front.

7.1 ROLE OF TURBULENCE IN FLAME ACCELERATION

One of the primary mechanisms driving flame acceleration is the interaction between the propagating flame front and turbulent flow structures. As the flame advances through a confined geometry, the thermal expansion of hot combustion products displaces the unburnt gases ahead of it. When this flow encounters obstacles or wall roughness, it generates turbulence within the unburnt region.

The resulting turbulence interacts with the flame front, causing it to wrinkle and distort, thereby increasing its total surface area. Since the rate of fuel consumption is directly proportional to the flame surface area, this leads to a rapid escalation in the overall heat release rate. The enhanced energy release strengthens the flow field, generating even greater turbulence—a self-reinforcing feedback loop that drives further acceleration of the flame. As a result, a smooth, laminar flame transforms into a spatially extended turbulent flame brush.

However, this process has intrinsic limits. Excessive turbulence can stretch and fragment the flame to the extent that local heat losses exceed the rate of heat generation, leading to flame quenching in localized regions. Consequently, the overall flame acceleration is governed by a delicate balance between turbulence-induced surface area enhancement and the suppressive effects of excessive flame strain and mixing.

7.2 SHOCK WAVE DYNAMICS AND FLAME–SHOCK INTERACTIONS

As the flame accelerates, it begins to behave like a piston, compressing the unburnt gas ahead of it and generating a field of compression waves. When the flame velocity approaches or exceeds the local speed of sound in the unburnt mixture, these compression waves can no longer escape freely and begin to coalesce into shock waves.

The interaction between the turbulent flame and these developing shocks plays a crucial role in the subsequent acceleration phase. The passage of a shock wave through the turbulent flame front amplifies the flame's wrinkling and enhances local reaction rates. At the same time, the shocks themselves may be strengthened by the energy released during combustion, resulting in a coupled flame–shock system.

The geometry of confinement significantly influences this dynamic. Reflections of precursor shocks from walls or obstacles can lead to shock focusing, where converging shock fronts produce localized regions of extreme pressure and temperature. These hot spots act as potential initiation sites for auto-ignition, creating favorable conditions for the transition to detonation. Converging surfaces, corners, and cavities are particularly effective at concentrating shock energy into small volumes, thereby triggering localized explosions within the accelerating flame field.

7.3 INFLUENCE OF OBSTACLES AND CONFINEMENT

In practical systems, obstacles and confinement are the dominant factors promoting DDT. Obstacles such as baffles, grids, or perforated plates serve as efficient turbulence generators, enhancing flame wrinkling and large-scale vortex formation. These vortical structures trap unburnt gas pockets that can combust violently, producing powerful jets of hot gas that drive the main flame front forward at an exponential rate.

The geometric characteristics of the obstacles—specifically their shape, spacing, and blockage ratio (BR)—play a decisive role in determining the rate of flame acceleration. Although higher blockage ratios typically correlate with faster acceleration due to stronger flow obstruction, the relationship is nonlinear and depends on the specific flow interactions produced.

Obstacles contribute to DDT through several complementary mechanisms:

- Generation of coherent flow structures (jets and vortices) that enhance large-scale flame motion.
- Modification of shock wave dynamics through reflection, focusing, and coalescence.
- Creation of preconditioned regions of elevated pressure and temperature that promote auto-ignition.

Confinement further augments these effects by preventing the dissipation of pressure waves. The accumulation and reflection of pressure transients lead to the formation of strong, interacting shock systems essential for detonation onset.

A key empirical relationship governing the likelihood of DDT involves the detonation cell size (λ)—a measure of the characteristic instability pattern within a propagating detonation. Experimental studies have shown that DDT is feasible only when the characteristic geometric dimension (such as tube diameter, d) exceeds the cell size, typically expressed as $d/\lambda \geq 1$. While this criterion provides a useful guideline, it is not universally sufficient, particularly in configurations with very high blockage ratios or extreme confinement.

7.4 HOT SPOT FORMATION AND AUTO-IGNITION: THE FINAL TRANSITION

The final stage of the DDT process occurs abruptly and is characterized by localized explosion events rather than a continuous increase in flame speed. The combined effects of turbulent mixing, shock compression, and shock focusing give rise to hot spots—small regions of unburnt gas that reach auto-ignition conditions prior to the arrival of the main flame front.

The near-isochoric combustion within these hot spots releases energy almost instantaneously, producing powerful blast waves. This phenomenon, described by Oppenheim as an "explosion within an explosion," plays a decisive role in triggering detonation. If a blast wave from one or more hot spots is sufficiently strong and expands into a region that has been preconditioned by earlier shocks, it can amplify into a self-sustaining detonation front that propagates through the remaining mixture.

Hot spots generally form in regions of high shear and compression—particularly near obstacles, walls, and locations where multiple shocks intersect. Their stochastic nature makes precise prediction difficult, but their formation marks the irreversible onset of detonation.

Chapter 8

PARAMETRIC ANALYSIS OF FACTORS INFLUENCING DDT

The occurrence of DDT depends on a complex interplay between chemical kinetics, initial thermodynamic conditions, and geometric configuration. Each of these parameters can significantly affect whether a deflagration transitions into a detonation.

8.1 CHEMICAL PROPERTIES OF THE COMBUSTIBLE MIXTURE

FUEL REACTIVITY

Fuels with short chemical induction times and high laminar flame speeds, such as hydrogen, acetylene, and ethylene, exhibit a higher propensity for DDT. Less reactive fuels like methane or propane generally require stronger confinement or obstacle-induced turbulence to achieve transition.

STOICHIOMETRY

Mixtures near stoichiometric proportions are most conducive to DDT, as they yield maximum energy release per unit mass and highest flame temperatures. This enhances expansion ratios and strengthens the compression waves that drive flame acceleration.

8.2 INITIAL THERMODYNAMIC CONDITIONS

PRESSURE

Higher initial pressures increase mixture density and reaction rates, reducing the critical distance and time required for DDT. Increased pressure also lowers the effective activation energy for ignition.

TEMPERATURE

Elevated initial temperatures promote faster chemical kinetics and reduce the extent of compressive heating needed to reach auto-ignition conditions. This accelerates the onset of DDT and strengthens the early pressure field.

8.3 GEOMETRIC AND BOUNDARY CONDITIONS

The geometric configuration of the system exerts a dominant influence on DDT behavior:

CONFINEMENT AND CHANNEL DIMENSIONS

Confinement is essential, as it enables pressure accumulation and shock reflection. In open geometries, energy dissipation through expansion waves generally prevents detonation formation. The available run-up distance and confinement length dictate whether the flame can reach the critical acceleration threshold.

OBSTACLE GEOMETRY

Obstacles strongly influence the flame's hydrodynamic evolution:

- **Blockage Ratio (BR):** Higher BR increases turbulence intensity and accelerates DDT, although excessive blockage can cause excessive pressure loss or quenching.
- **Shape:** Sharp-edged or bluff obstacles generate stronger turbulence, while streamlined ones promote smoother flow and slower transition.
- **Spacing:** Optimal spacing ensures effective interaction of vortex structures and shock reflections, maintaining continuous flame acceleration.

In summary, the deflagration-to-detonation transition represents a synergistic coupling of turbulence, shock dynamics, and chemical reactivity. While numerous parameters influence the process, it fundamentally relies on achieving conditions where localized auto-ignition events can occur in a confined, highly turbulent, and compressively preconditioned environment.

Chapter 9

COMPUTATIONAL FLUID DYNAMICS (CFD) SIMULATION SETUP FOR DEFLAGRATION-TO-DETONATION TRANSITION (DDT)

This section details the numerical methodology employed for the simulation of the Deflagration-to-Detonation Transition (DDT) phenomenon in a hydrogen-air mixture using the commercial CFD software, ANSYS Fluent.

9.1 GEOMETRY AND COMPUTATIONAL MESH

The computational domain was configured as a two-dimensional, axisymmetric representation of an obstructed tube.

DOMAIN DIMENSIONS

The model consists of a rectangular tube with a total length of 1.0 meter and a width of 50 mm.

OBSTACLES

To promote flame acceleration and the onset of DDT, rectangular obstacles were placed along the tube wall. Each obstacle has a height and width of 8 mm, spaced 42 mm apart from each other. This configuration was specifically designed to replicate the blockage ratio of a 400 mm spiral used in a physical Pulsed Detonation Rocket Engine (PDRE) tube, ensuring the simulation is representative of relevant experimental conditions.

MESHING

The computational mesh was generated using ANSYS Mechanical. A structured grid with a uniform element size of 1 mm was applied throughout the domain. This resulted in a high-resolution mesh comprising approximately 48,800 elements, which is sufficiently fine to capture the complex flow physics, flame propagation, and shock wave interactions.

9.2 SOLVER AND PHYSICAL MODELS

The simulation was performed using the ANSYS Fluent solver with a setup tailored for high-speed, compressible, reactive flows.

SOLVER TYPE

A Density-Based solver was selected, as it is robust and more accurate for simulations involving high Mach numbers and significant compressibility effects, which are characteristic of detonation events. The analysis was set to Transient to capture the time-evolution of the DDT process.

ENERGY EQUATION

The energy equation was enabled to solve for temperature variations resulting from chemical heat release and compressible effects.

9.2.1 TURBULENCE MODELING: DDES

The Delayed Detached Eddy Simulation (DDES) model was chosen to model turbulence.
DDES Model Explained: DDES is a hybrid Reynolds-Averaged Navier-Stokes (RANS) and Large Eddy Simulation (LES) model. It functions as a RANS model (like SST k-ω) within the boundary layer, where flow is stable and attached to surfaces, providing computational efficiency. In regions of separated flow away from walls, it switches to an LES model, resolving the large-scale turbulent eddies directly. The "Delayed" aspect prevents premature switching to LES, ensuring the RANS model remains active where most effective. This hybrid approach balances computational cost with the accuracy required to capture turbulent flame dynamics.

9.2.2 COMBUSTION MODELING: SPECIES TRANSPORT

To simulate the chemical reactions, the Species Transport model was utilized.
Species Transport & Finite-Rate Model Explained: The Species Transport model solves individual conservation equations for each chemical species present in the mixture (e.g., H_2, O_2, N_2, H_2O), allowing tracking of local mass fractions over time. Reaction source terms were calculated using the Finite-Rate model, which computes reaction rates directly from Arrhenius expressions as a function of species concentration, temperature, and rate constants. This is essential for DDT simulations to capture the detailed chemical kinetics governing the transition from deflagration to detonation.
Fluid Material: The working fluid was defined as a Hydrogen-Air mixture, with the corresponding species and reaction mechanisms defined within the model.

9.3 BOUNDARY CONDITIONS AND OPERATING CONDITIONS

The following boundary conditions were applied to the computational domain:

WALLS

The tube and obstacle walls were defined with a No-Slip shear condition (fluid velocity at the wall is zero) and set as Adiabatic (no heat transfer).

OUTLET

The open end of the tube was defined as a Pressure Outlet set to standard atmospheric pressure (101,325 Pa).

OPERATING CONDITIONS

The operating pressure was set to 0 Pa, ensuring all reported pressures are absolute, which is crucial for analyzing high-pressure shock waves in detonation.

9.4 NUMERICAL SOLUTION SCHEME

The following numerical methods were employed for accuracy and stability:

FLUX TYPE

Advection Upstream Splitting Method (AUSM).

SPATIAL DISCRETIZATION

- **Gradient:** Least Squares Cell-Based.
- **Flow Variables (Pressure, Density, Momentum, etc.):** Second Order Upwind scheme.
- **Transient Formulation:** Second Order Implicit scheme.

SOLUTION CONTROLS

The Courant Number (CFL) was set to 0.5 to maintain numerical stability, limiting the distance the flow travels across a mesh cell in a single time step, which is critical for resolving steep gradients associated with shocks and detonations.

9.5 INITIALIZATION AND RUN CALCULATION

INITIALIZATION

The flow field was first initialized using the Hybrid Initialization method. Patching was then applied to a small, user-defined region, setting high pressure (10 atm) and high temperature (1000 K) to act as the ignition source.

RUN CALCULATION

The transient simulation was executed with the following parameters:

- **Time Step Size (Δt):** very small (to resolve rapid detonation timescales).
- **Number of Time Steps:** 1000.
- **Iterations per Time Step:** Maximum of 20 to ensure convergence.

9.6 POST-PROCESSING

For data analysis and visualization, the following were configured:

XY PLOT

A centerline was defined along the tube to generate XY plots of Pressure vs. Position, tracking shock and detonation waves.

CONTOURS

Contour plots for Pressure, Velocity, Density, and Mach Number were generated to visualize flow field, flame front, and shock structures at various time instances.

Chapter 10

RESULTS AND DISCUSSION

This section presents the results obtained from the transient CFD simulation of the Deflagration-to-Detonation Transition (DDT) process. The analysis focuses on validating the formation of a stable detonation wave by comparing the simulation data against established theoretical models, primarily the Chapman–Jouguet (CJ) theory and the Zel'dovich–von Neumann–Döring (ZND) model.

10.1 QUALITATIVE ANALYSIS OF THE FLOW FIELD

The simulation successfully captured the transition from a slow-moving deflagration flame to a high-speed, self-sustaining detonation wave. Visual analysis of the generated contour plots provides a clear depiction of the detonation structure as it propagates through the tube.

PRESSURE CONTOURS

The pressure contour reveals the characteristic feature of a detonation wave: an extremely sharp, abrupt jump in pressure at the leading shock front, known as the von Neumann spike. This is immediately followed by a region of decreasing pressure corresponding to the reaction zone, eventually settling to a stable post-detonation pressure.

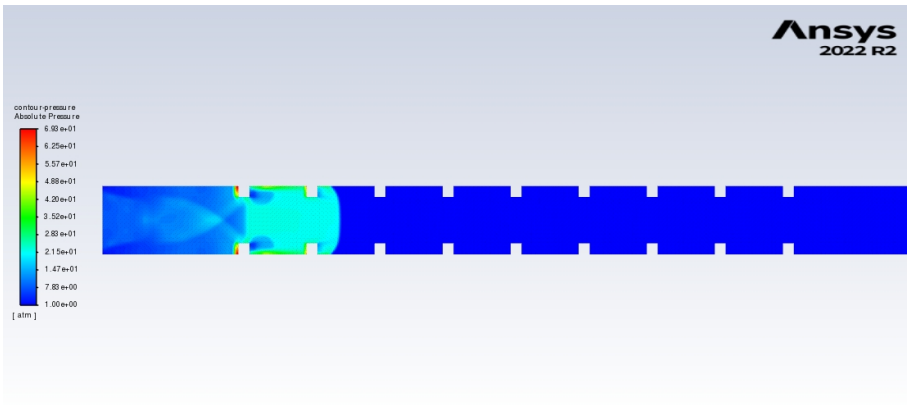

Figure 10.1: Pressure Contour

TEMPERATURE AND DENSITY CONTOURS

The temperature and density contours exhibit an inverse relationship, as expected. The leading shock front instantaneously compresses the unburnt gas, causing a sharp increase in both temperature and density. Following this shock, the exothermic energy release in the reaction zone causes a further, massive increase in temperature and a corresponding sharp decrease in density as the hot product gases expand.

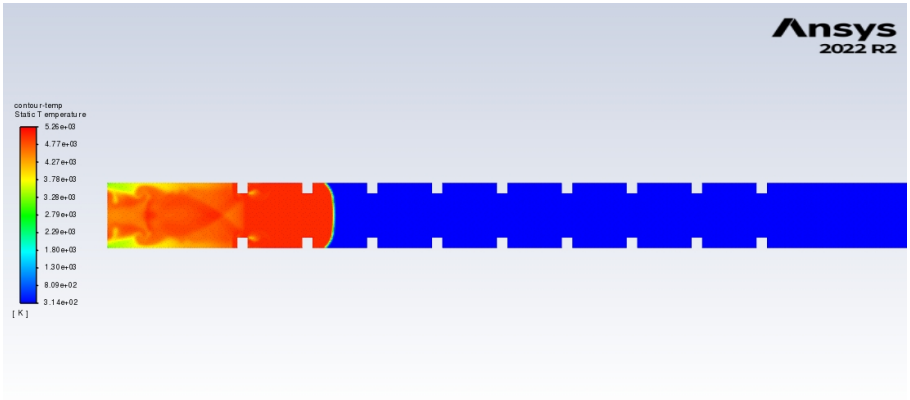

Figure 10.2: Temperature Contour

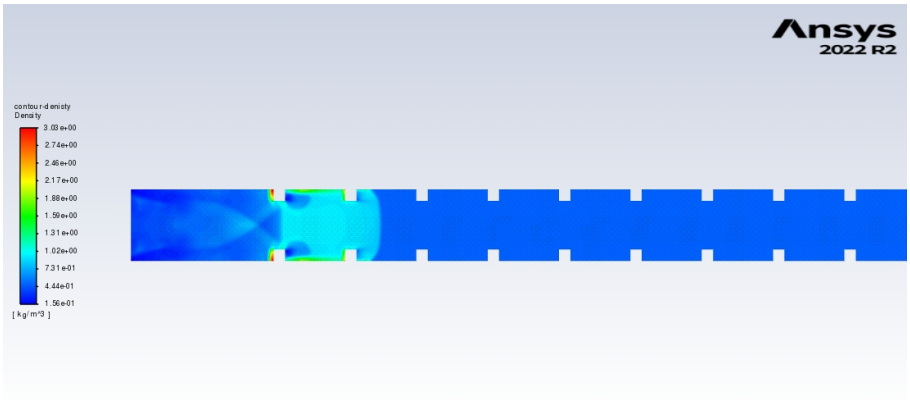

Figure 10.3: Density Contour

MACH NUMBER CONTOURS

The Mach number plot is critical for verifying the CJ condition. It clearly shows the supersonic nature of the propagating wave. Crucially, the flow immediately behind the reaction zone, in the post-detonation region, is observed to be sonic ($M \approx 1$), satisfying the fundamental tenet of the Chapman–Jouguet condition.

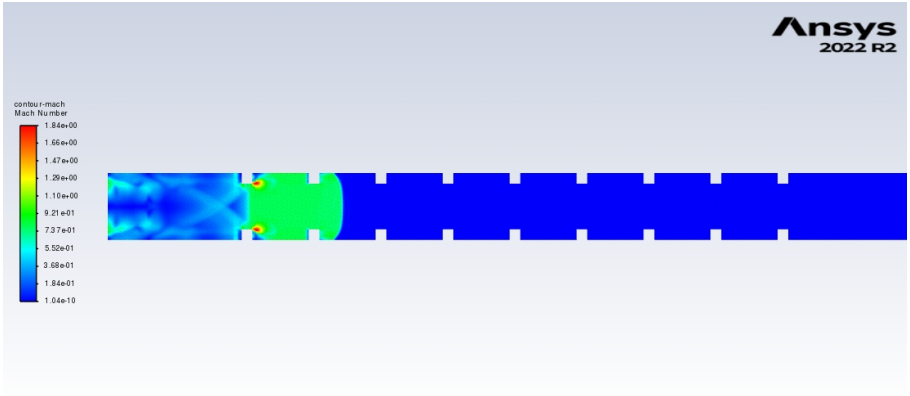

Figure 10.4: Mach Number Contour

10.2 QUANTITATIVE VERIFICATION AGAINST DETONATION THEORY

To quantitatively validate the simulation, key parameters of the stable detonation wave were extracted and compared with theoretical values derived from the Chapman–Jouguet theory for the stoichiometric hydrogen–air mixture. The pressure profile along the tube centerline provides the most precise data for this comparison.

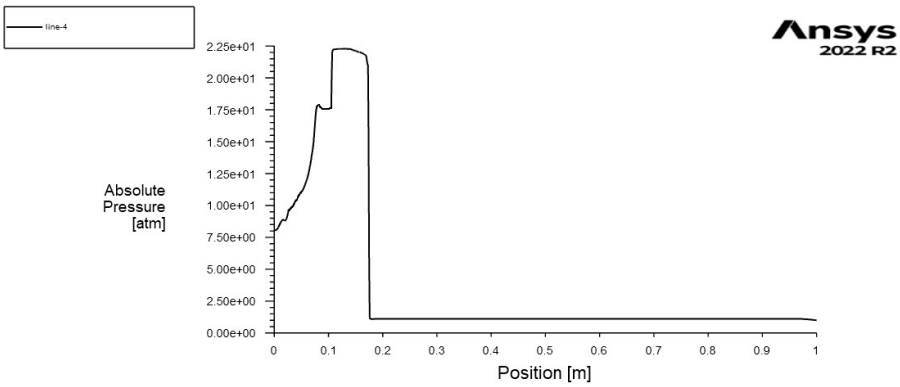

Figure 10.5: Pressure Plot

- **Peak Pressure (Von Neumann Spike):** The ZND model predicts a sharp pressure spike immediately behind the shock front. The theoretical peak pressure for this mixture is approximately 25 atm. The simulation captured a peak pressure of 22.5 atm, showing good agreement.
- **Chapman–Jouguet (CJ) Pressure:** The theoretical CJ pressure is 15 atm. The simulation shows a stable post-detonation pressure of 17 atm.
- **Detonation Mach Number:** The theoretical CJ Mach number is 4.8. The Mach number in the simulation was found to be 4.4.

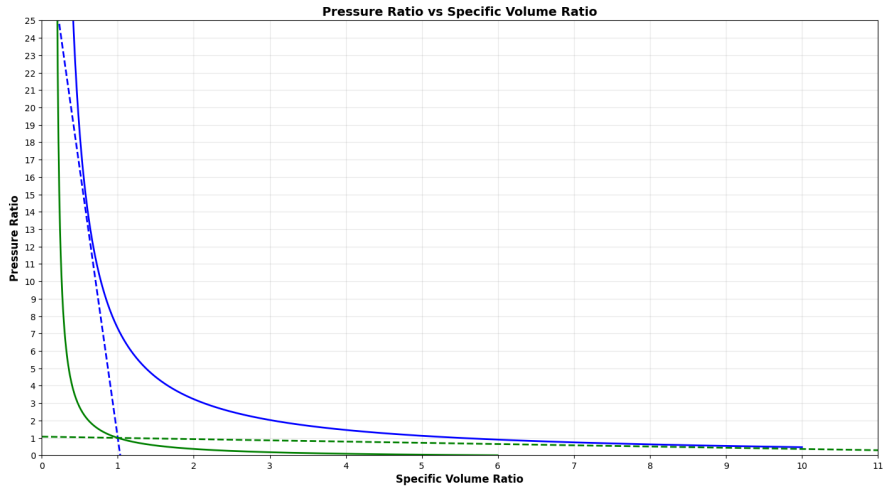

Figure 10.6: Rayleigh Hugoniot Curve

Table 10.1: Comparison of simulated detonation data with experimental data

Parameter	Theoretical Value (CJ/ZND)	Simulated Value	Percent Error
Peak Pressure (P_{VN})	25 atm	22.5 atm	-10%
CJ Pressure (P_{CJ})	15 atm	17 atm	+13%
Mach Number	4.8	4.5	-6.2%

A summary of this comparison is presented in Table 10.1.
The results show strong correlation with theoretical predictions. The simulated detonation Mach number and peak pressure are within reasonable margins of error. The discrepancy in CJ pressure (+13%) may be attributed to factors such as chemical kinetics modeling, mesh resolution, or turbulence modeling. Overall, the quantitative agreement confirms the successful modeling of a detonation wave consistent with CJ and ZND conditions.

10.3 VERIFICATION OF THE COUPLED SHOCK–REACTION STRUCTURE

A defining feature of a detonation wave, as described by the ZND model, is the tight coupling between the leading shock front and the subsequent chemical reaction zone. To confirm this, the species concentrations of the primary reactant (H_2) and a key product (H_2O) were analyzed.
The plots and contours of species mass fractions demonstrate this coupling:

- At the shock front, the concentration of H_2 remains high while the concentration of H_2O is zero.
- Immediately following the shock, over a finite distance known as the reaction zone, H_2 rapidly decreases, while H_2O correspondingly increases, signifying ongoing chem-

ical reactions.

- Once the reaction is complete, H_2 approaches zero, and H_2O reaches its stable post-detonation value.

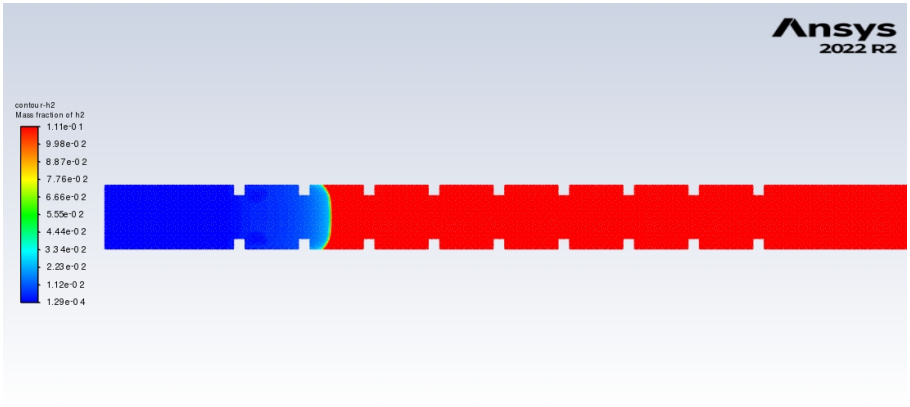

Figure 10.7: Mass Fraction of H2 Contour

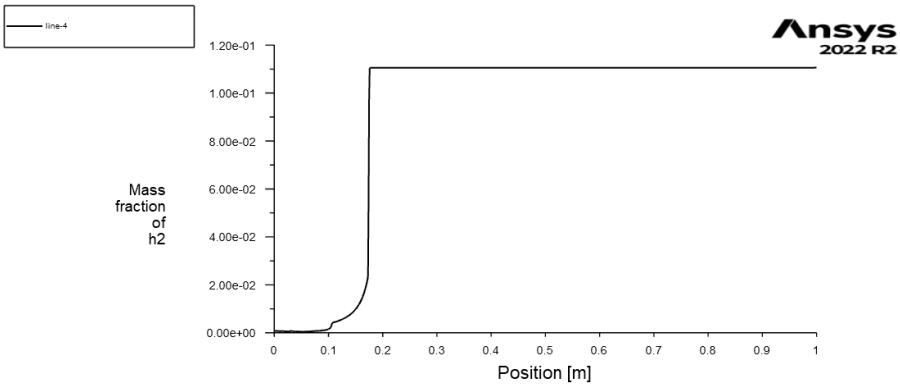

Figure 10.8: Mass Fraction of H2 Plot

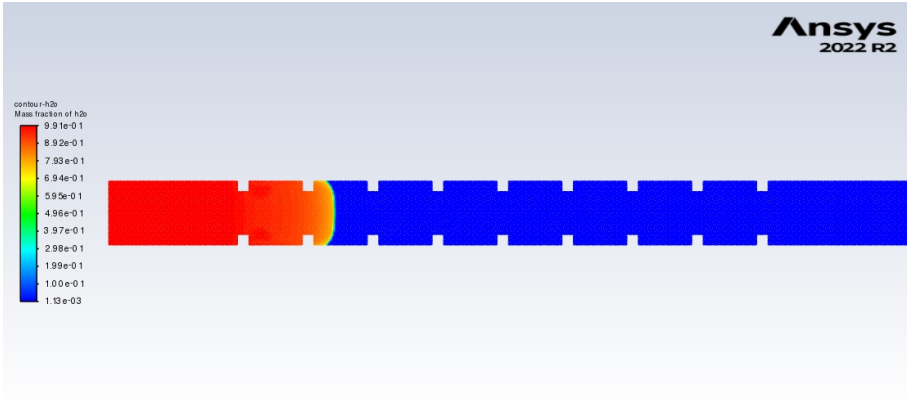

Figure 10.9: Mass Fraction of H2O Contour

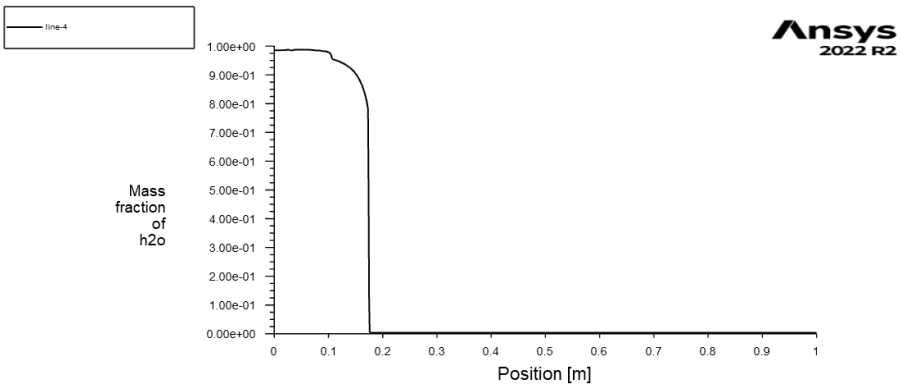

Figure 10.10: Mass Fraction of H2O Plot

This rapid conversion of reactants to products in a thin zone immediately behind the leading shock confirms the existence of the ZND structure, providing conclusive evidence that the simulated event is a true detonation and not a decoupled shock and deflagration.

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